



Nonparametric estimation of matching efficiency and elasticity in a spot gig work platform: 2019–2023[☆]

Hayato Kanayama^a,^{*}, Suguru Otani^b

^a Graduate School of Economics, Waseda University, Japan

^b Market Design Center, Department of Economics, The University of Tokyo, Japan

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ABSTRACT

This paper provides new evidence on spot gig work platforms for individuals seeking flexible, short-term jobs with minimal educational or experience requirements in Japan. Using proprietary data from Timee, a private matching platform, the study analyzes trends in active users, vacancies, hires, and labor market tightness, compared to part-time data from Hello Work, a public employment service. Applying a nonparametric approach, it finds that the private platform exhibits substantially higher matching efficiency, especially after 2022. Elasticities also differ across platforms: for Hello Work, the user elasticity fluctuates around 0.3–0.5, while the vacancy elasticity ranges roughly from 0.4 to slightly above 1.0; for the private platform, the user elasticity remains around 0.2–0.3, while the vacancy elasticity ranges from 0.7 to 1.1. At the prefecture level, the three prefectures exhibit broadly similar movements early in the sample, followed by divergence and partial re-convergence later on, while elasticities remain stable and similar across regions. These results reveal how digital platforms reshape job matching dynamics relative to traditional systems.

1. Introduction

The gig economy, characterized by temporary, contract, and freelance online jobs rather than permanent positions, has experienced rapid growth in recent years. An increasing proportion of freelance workers are now matched with customers through online platforms. By analyzing the number of open projects and tasks on a sample of such platforms, Kässi and Lehdonvirta (2018) find that the demand for online gig work increased by approximately 21% from 2016 to 2018. According to *Freelance Forward 2023* reported by Upwork, the share of professionals freelancing increased to nearly 64 million Americans, making up 38% of the U.S. workforce in 2023.¹ Despite the significance of the gig economy, research directly analyzing the structure and efficiency of gig labor markets remains limited, especially in comparison to traditional labor markets facilitated by public job search platforms.

This paper contributes to the literature by providing the first systematic analysis of a spot gig worker labor market, distinguishing it from traditional labor markets and other segments of the gig economy.

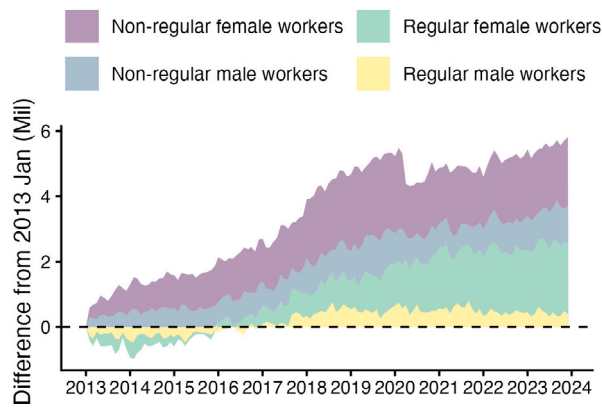
We focus on labor market matching efficiency and elasticities within spot work platforms that cater to workers who want to work as an on-demand labor force or are unemployed, seeking flexible, short-term jobs requiring minimal education or experience. Using proprietary aggregate data from a private online spot work matching platform, Timee, in Japan, we document key trends and compare these to public-sector employment services. The significance of spot work markets has expanded considerably from 2013 to 2024, as shown in Panels (a) and (b) of Fig. 1. First, we analyze trends in core aggregate variables, including the number of Timee users, vacancies, hires, and labor market tightness (i.e., the ratio of vacancies to Timee users). To benchmark our findings against the public-sector labor market, we incorporate data from the “Report on Employment Service” (*Shokugyo Antei Gyomu Tokei*), which provides insights into part-time employment trends facilitated through Hello Work, Japan’s public employment service. This comparison sheds light on the structural differences not only between private and public job-matching mechanisms but also between gig and standard jobs in Japan.

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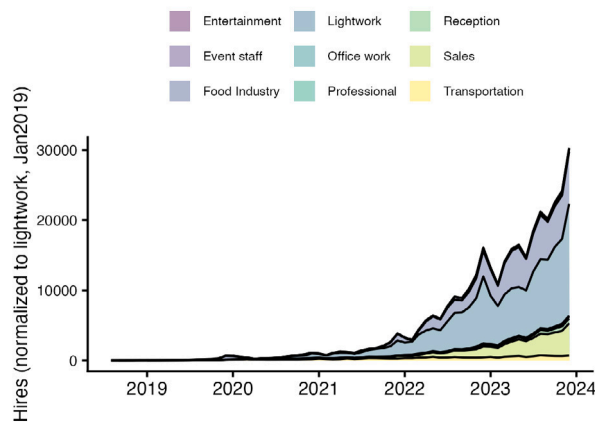
^{*} Corresponding author.

E-mail addresses: ha13yato@toki.waseda.jp (H. Kanayama), suguru.otani@e.u-tokyo.ac.jp (S. Otani).

¹ <https://www.upwork.com/research/freelance-forward-2023-research-report>: Accessed on September 19, 2024.



(a) Hires, Labor force survey



(b) Hires, Spot work platform

Fig. 1. Changes of employment. National vs Spot work platform.

Note: Panel (a) illustrates the time series of job changes in Japan, standardized to January 2013. The colored areas represent changes within specific worker groups. This graph is derived from data provided by the Economic Survey conducted by the Japanese Cabinet Office (details available at <https://www5.cao.go.jp/keizai3/2024/0802wp-keizai/setsumei-e2024.pdf>, accessed September 19, 2024). Panel (b) presents the time series of hires on the platform, standardized to January 2019 for confidentiality purposes. The vertical axis represents the relative number of hires compared to January 2019. The filled areas depict the number of matched spot workers across various job categories as defined by the respective companies. See the survey of Miyamoto (2025) for an overview of labor markets in Japan.

First, we document the rapid expansion of the spot labor market on the private platform between December 2019 and December 2023. The framework follows (Cullen and Farronato, 2021) which studies a more housework-specific spot job platform, TaskRabbit. The number of registered Timee users, predominantly unemployed individuals actively seeking spot work, exhibits a gradual increase, while the number of vacancies surges—particularly after 2022—leading to a rise in labor market tightness. Hiring activity escalates significantly around 2020, reflecting a sharp rise in successful job matches. Additionally, the job finding rate surpasses 1.0, indicating that active users frequently secure multiple spot jobs per month. Meanwhile, the worker finding rate remains stable at around 0.8, showing that 80% of spot jobs are successfully filled, reinforcing the platform's effectiveness in facilitating rapid job matching and featuring the matching frequency of the spot gig work platform.

Second, we estimate the matching function and recover matching efficiency and elasticities using the novel nonparametric approach developed by Lange and Papageorgiou (2020). The approach is applicable to a wide range of matching markets, including full-time labor and marriage markets (Otani, 2024, 2025b,a). Our results reveal clear distinctions between public and private job-matching mechanisms. For Hello Work, matching efficiency remains relatively stable, fluctuating within a narrow range around the baseline. In contrast, the private platform exhibits a substantial rise in matching efficiency, peaking around May 2023, coinciding with Japan's peak holiday season, Golden Week. Elasticities also differ: for Hello Work, the elasticity with respect to users fluctuates around 0.3 to 0.5, while the elasticity with respect to vacancies ranges roughly from 0.4 to slightly above 1.0. For the private platform, the elasticity with respect to users remains relatively low, around 0.2 to 0.3, whereas vacancy elasticity ranges from 0.7 to 1.1, indicating stronger responsiveness to job demand than to worker inflows.

These comparisons should be interpreted based on differences in the measurement of variables and the composition of workers, vacancies, and hires across the two datasets, as well as the substantive difference between more standard job matching through Hello Work and spot gig work on an app-based platform. Estimating a matching function for both Timee and Hello Work is conceptually appropriate because both settings involve hires, job seekers, and vacancies, but differences in how these objects are measured and constituted affect how the estimated efficiency and elasticities should be interpreted. Accordingly, differences between the two platforms should be understood as reflecting not only differences in matching mechanisms, but also differences in the underlying data structure and in the type of jobs being matched. At the same time, this comparison provides broader insight into how gig work can be incorporated into search-and-matching models and how app-based spot labor markets differ from more traditional matching environments.

Lastly, we examine prefecture-level trends, focusing on Aichi, Osaka, and Tokyo—Japan's three largest metropolitan areas. This comparison is intended to assess whether the platform exhibits similar matching performance across major regions, thereby shedding light on geographic heterogeneity in spot-work matching on an app-based platform. We focus on these prefectures for two reasons. First, at the prefecture level, many smaller prefectures are too thin, especially early in the sample, for spot-work matching to form a meaningful market-level series. Second, Kanayama et al. (2026) documents that Timee first launched in major metropolitan prefectures such as Tokyo, Osaka, and Aichi, which allows us to trace platform development from the early phase onward. The three prefectures exhibit broadly similar movements early in the sample, followed by divergence and partial re-convergence later on. Elasticities related to both users and vacancies remain stable and similar in magnitude across regions, suggesting limited geographic heterogeneity in matching responsiveness on the platform.

This study lays the foundation for understanding the economic mechanisms governing spot gig work, extending labor economics research beyond traditional and even broader gig economy studies. By shedding light on the distinct characteristics of spot work platforms and their efficiency in labor market matching, we offer novel insights into a rapidly evolving segment of the labor market.

1.1. Related literature

First, this paper contributes to the empirical literature on estimating the matching function, a fundamental component in macroeconomic models. We examine the trend of matching efficiency in Japanese labor markets via an online spot work matching platform nonparametrically using a novel approach (Lange and Papageorgiou, 2020), which shows how to nonparametrically identify the matching function and estimate the matching function allowing for unobserved matching efficacy, without imposing the usual independence assumption between matching

efficiency and search on either side of the labor market, allowing for multiple types of jobseekers. Lange and Papageorgiou (2020) highlight positive correlations between efficiency and market structure such as tightness and so on, which induces a positive bias in the estimates of the vacancy elasticity whenever unobserved matching efficacy is not controlled for, as is the case in the traditional Cobb–Douglas matching function with constant elasticity parameters.²

In the context of Japanese labor markets, Otani (2024) estimates matching efficiency and mismatch in the off-the-job search administrative worker-vacancy-matching platform via Public Employment Security Offices, known as Hello Work. The paper finds a declining trend in matching efficiency, consistent with decreasing job and worker finding rates. The implied match elasticity with respect to unemployment is 0.5–0.9, whereas the implied match elasticity with respect to vacancies is between –0.4 and 0.4. Applying the same approach to proprietary data, Otani (2025b) estimates matching efficiency and elasticity in the on-the-job search scouting platform for high-skill employed workers, operated by a private firm in Japan, and compares these with those for Hello Work full-time workers. This paper complements the above findings. A series of papers overview the Japanese labor markets in the 2010s and 2020s and provide empirical evidence on matching efficiency, elasticity, and mismatch.

Second, our paper contributes to a nascent literature on alternative work arrangements in labor economics (Mas and Pallais, 2020), which are closely related to spot work markets. The key difference between spot work and classical work is the frequency and flexibility of labor supply and demand. This paper is the first to estimate matching efficiency and elasticity to the number of registered workers and spot vacancies on the online spot work platform highlighting the key difference from the public part-time worker-job matching platform. Alternative work arrangements based on spot platforms are growing rapidly in labor markets. Katz and Krueger (2019) focus on alternative work arrangements defined as temporary help agency workers, on-call workers, contract workers, and independent contractors in the United States from 2005 to 2015. They provide evidence of an increase in these workers but online platform workers, such as “Uber” and “Task Rabbit” represented only 0.5% of all workers in 2015. By contrast, Kassi and Lehdonvirta (2018) show that the demand for online labor markets increased by 21% from 2016 to 2018.³

Focusing on the important feature of search and matching environment, some transportation industries such as taxi (Frechette et al., 2019; Buchholz, 2022; Lehe and Pandey, 2022), ride-sharing like Uber (Guda and Subramanian, 2019; Rosaia, 2020; Castillo, 2023; Castillo et al., 2024), bulk shipping (Brancaccio et al., 2020, 2023) are well

studied. These studies describe models to explain why inefficiencies occur in the search and matching of sellers and buyers. However, to our knowledge, there is less literature about the overall features in the spot labor markets where the labor force on customer service, sales, and cleaning service is demanded and this is the first article to investigate the matching efficiency in these markets.

Third, this paper contributes to the growing literature on online job search platforms. The analysis of job matching within actual market institutions has gained prominence with the increasing availability of data from job advertising platforms (Autor, 2019).⁴ Much of the literature examines application-level or vacancy-level behavior to capture search behavior and wage elasticity. For instance, Faberman and Kudlyak (2019) use proprietary application-level data from an online job search engine to investigate the relationship between search intensity and search duration, focusing on lower-skill, hourly jobs for both employed and unemployed workers. Similarly, Kambayashi et al. (2025) estimate elasticities of application, interview attendance, and offer acceptance with respect to posted wages using detailed process-level data for high-skill workers and firms on private job search platforms in Japan. In contrast, this paper takes a broader macro-level view, evaluating the efficiency of the private matching platform itself. To our knowledge, this is the first paper to estimate matching efficiency and elasticity in an online spot work matching platform, providing relatively longer-term insights into private online job search trends and complementing the micro-level studies mentioned above.

The rest of the paper is organized as follows. Section 2 outlines the platform and the Hello Work data. Section 3 introduces our model of nonparametric aggregate matching function. Section 4 describes our method of estimation. Section 5 reports the main results. Section 6 concludes.

2. Data

2.1. Data source

First, we use the Report on Employment Service (*Shokugyo Antei Gyomu Tokei*) for the month-level aggregate data from December 2019 to December 2023 to capture trends in matches between unemployed workers and vacancies via the public employment service. These datasets include the number of job openings, job seekers, and successful job placements, primarily sourced from the Ministry of Health, Labour and Welfare (MHLW) of Japan, which publishes monthly reports and statistical data on the Public Employment Security Office, commonly known as Hello Work. These series are not seasonally adjusted, because our objective is to estimate the matching function flexibly in a nonparametric way rather than impose seasonal effects through fixed effects. Hello Work is a government-operated institution in Japan that provides job seekers with employment counseling, job placement services, and vocational training, playing a critical role in Japan’s labor market. The data is often used as in Kano and Ohta (2005), Kambayashi and Ueno (2006), Sasaki (2008), Kambayashi (2013), Higashi (2018), and Kawata and Sato (2021). In particular, Shibata (2020) estimates the traditional Cobb–Douglas matching function, whereas (Otani, 2024) estimates the nonparametric matching function. In this study, we focus on part-time workers in Hello Work for a reasonable comparison. The sample period is chosen to match the timeframe of the platform data below.

Second, we use proprietary data from Timee, a private company that operates a spot-worker matching platform in Japan, to analyze trends

² Using administrative data of users and vacancies, Petrongolo and Pissarides (2001) summarize the early aggregate studies based on a Cobb–Douglas matching function with the flow of hires on the left-hand side and the stock of unemployment and job vacancies on the right-hand side. In short, the match elasticity with respect to unemployment is in the range 0.5–0.7. In the context of Japanese labor markets, Otani (2024) uses the nonparametric approach and updates the existing findings reported in Kano and Ohta (2005), Kambayashi and Ueno (2006), Sasaki (2008), and Higashi (2018) which use the traditional Cobb–Douglas matching function with geographical and occupational category fixed effects to capture geographical and occupational heterogeneity, which are also useful for comparison with other countries’ results reported in Bernstein et al. (2022) and Petrongolo and Pissarides (2001).

³ Utilizing the unique features, empirical literature characterizes the relationship between labor supply and compensation scheme (Chen et al., 2019; Angrist et al., 2021; Hall et al., 2021; Butschek et al., 2022) and gender wage gap (Cullen et al., 2018; Cook et al., 2021; Adams et al., 2025). Angrist et al. (2021) have the randomized experiment using the “Uber” service to compare the value of the proportional compensation scheme offered by ride-share companies with taxi compensation. Butschek et al. (2022) implement a field experiment in the gig economy to confirm workers’ intrinsic motivation affects the relationship between performance and the compensation scheme.

⁴ For example, studies such as (Kuhn and Skuterud, 2004), Kuhn and Mansour (2014), and Kroft and Pope (2014) use data focused solely on worker status, while papers like (Kuhn and Shen, 2013), Hershbein and Kahn (2018), Brown and Matsa (2016), and Azar et al. (2020) focus exclusively on vacancy data. Other studies, such as (Banfi and Villena-Roldan, 2019), Marinescu and Rathelot (2018), Marinescu and Wolthoff (2020), and Azar et al. (2022) utilize data that captures both worker and vacancy information.

in matches between part-time spot workers and vacancies. Unlike Hello Work, the platform operates as an on-demand staffing service designed to connect businesses with temporary workers for short-term jobs. Its primary users are workers seeking flexible, task-based employment rather than long-term positions, including unemployed individuals and those who want to work on demand. Workers can register on the app for free and immediately access job postings from various companies, while the platform streamlines the matching process for shifts across industries such as food, retail, and logistics.⁵

The platform's business model allows workers to browse and accept jobs without the need for a traditional hiring process. Instead of charging a subscription fee, the platform earns revenue through fees paid by the businesses utilizing the service. This arrangement benefits both companies and workers by offering flexible, short-term employment opportunities while avoiding the formalities and commitments of long-term contracts. The platform's simplicity and immediacy are its main draws for workers looking to fit employment around their schedules.

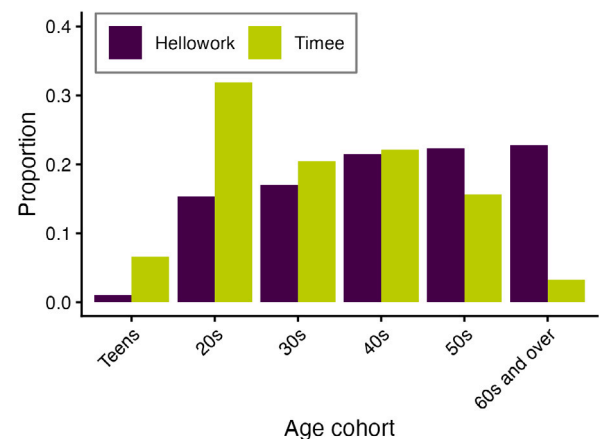
Several remarks are noteworthy. First, active job seekers on the platform are defined as registered workers who have activated the Timee app within a given month. Some of these active users may only complete registration in that month without engaging in actual job search activity. Therefore, active users on Timee do not correspond exactly to unemployed job seekers in the Hello Work data.⁶ Second, unlike Hello Work, where job seekers must enter part-time contracts for a set period and then leave the platform, active job seekers on Timee can match with multiple spot vacancies in a month. Third, some registered workers may use the platform for spot work as a side job. Because our analysis focuses on within-platform search and matching, we do not distinguish such users separately in the analysis.

Panel (b) in Fig. 1 shows a diverse range of job categories, with Entertainment, Food Industry, and Office Work occupying significant shares. Other categories like Event Staff, Light Work, and Professional roles also contribute to the platform's job distribution, albeit to a lesser extent. This surge suggests a growing reliance on the spot work platform for flexible employment opportunities, particularly in sectors like Entertainment and Food Industry. The steady rise across multiple categories implies an expanding spot labor market, reflecting a broadening acceptance and utilization of on-demand work arrangements in Japan.

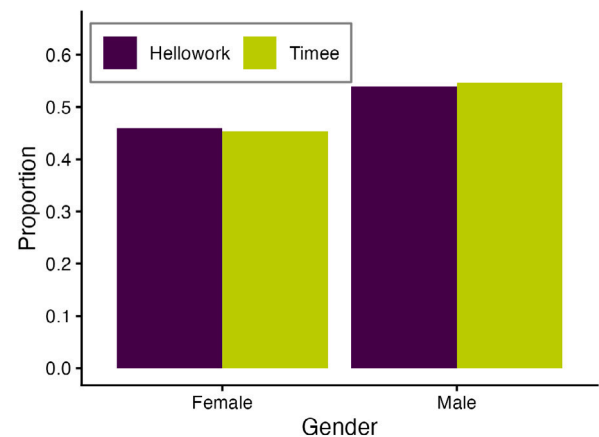
2.2. Age and gender comparison

To explore the difference between Hello Work and the private spot work platform, we compare user characteristics across the two platforms. This comparison helps illustrate qualitative differences in the nature of search and matching across the two platforms.

Fig. 2 presents the age-cohort and gender composition of hires across the two platforms in 2023. Panel (a) of Fig. 2 presents the age-cohort distribution of hires in 2023. The purple bars denote the hiring shares on Hello Work, while the green bars denote those on the private platform. The age distribution of hires differs markedly between the two platforms. On the private platform, workers in their twenties account for the largest share of hires, at around 30 percent, whereas workers in their fifties and those aged 60 or above account for relatively smaller shares. By contrast, on the public platform, workers in their



(a) Age cohort comparison, 2023



(b) Gender comparison, 2023

Fig. 2. Age cohort and gender comparison between Hello Work and the platform.

Note: Panel (a) shows the age-cohort distribution of hires in 2023. Panel (b) shows the gender distribution of hires in 2023.

twenties and thirties represent relatively smaller shares, while workers aged 40 and above each account for more than 20 percent of hires in a relatively stable manner. These patterns suggest that the two platforms serve different segments of the labor market: the private platform is more strongly oriented toward younger workers, whereas the public platform covers a larger share of middle-aged and older workers.

Panel (b) of Fig. 2 presents the distribution of hires by gender in 2023. The purple bars denote the hiring shares on Hello Work, while the green bars denote those on the private platform. In contrast to the age distribution, the gender composition of hires is broadly similar across the two platforms. In both the private and public platforms, male hires account for more than 50 percent of total hires, whereas female hires account for less than 50 percent. This pattern suggests that the two platforms do not differ substantially in terms of the gender composition of their users.

2.3. Trend comparison

Fig. 3 provides a comparative analysis of labor market dynamics between Hello Work (left panel) and the private spot work platform (right panel) from December 2019 to December 2023.

First, we discuss the left panel of Hello Work for part-time workers and jobs. In panel (a), the number of unemployed individuals remains broadly stable over the observed period, hovering around 0.8 million.

⁵ Using the same data, Kanayama et al. (2026) study the effect of the minimum wage on spot employment on the platform using a Difference-in-Differences approach. See Kanayama et al. (2026) for more details about the platform and vacancy-side variables. In addition, Sekiya et al. (2026) conduct a field experiment to investigate the effect of their proposed algorithm on job-favoring and application behavior.

⁶ In the Hello Work system, job-seeker registration remains valid, in principle, until the end of the second month following the month of registration, and it can be extended on a monthly basis when the job seeker receives employment services or engages in job search activity.

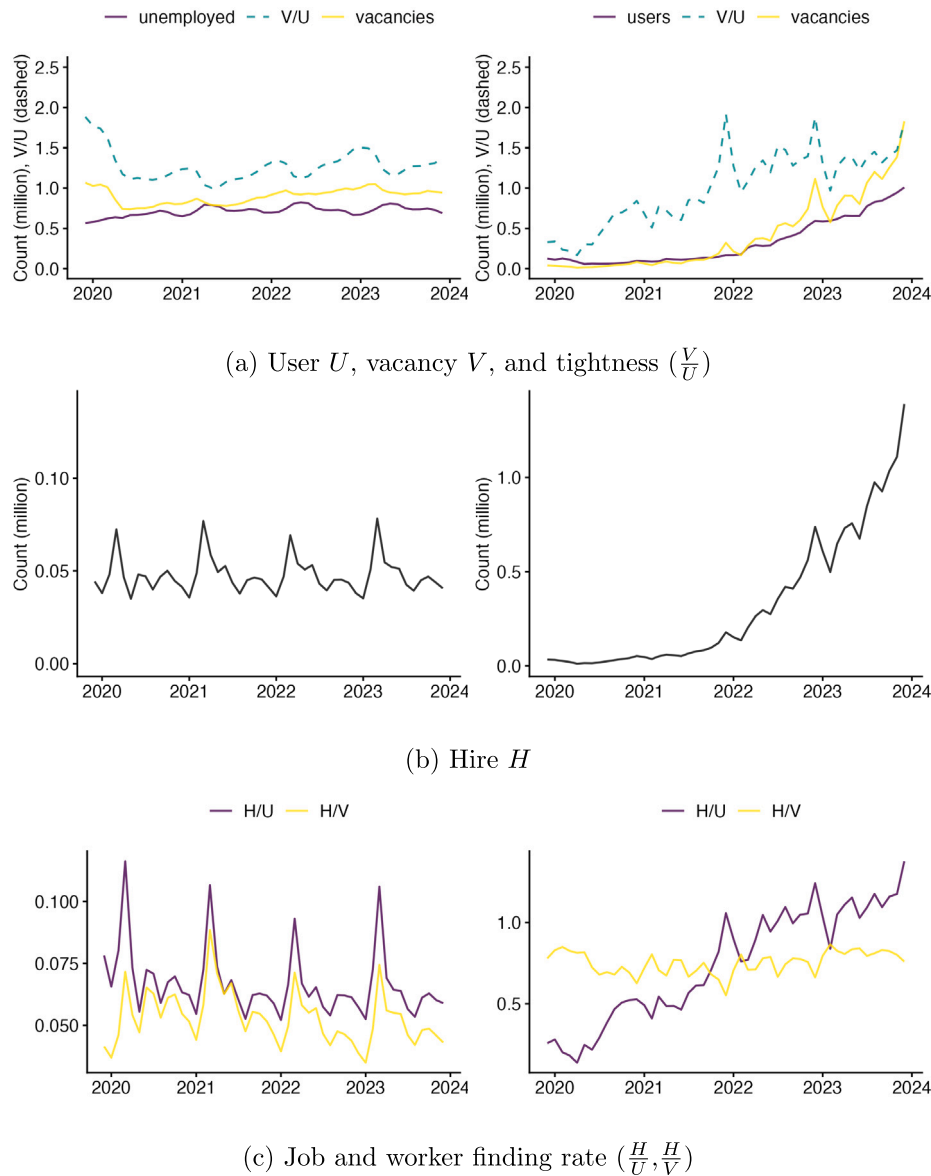


Fig. 3. Trends of key variables: Hello Work part-time (left) vs platform (right) 2019–2023.

Vacancies and labor market tightness (V/U) decline in the first half of 2020, likely reflecting the disruption caused by the COVID-19 pandemic, but remain broadly stable thereafter without a clear recovery to the pre-pandemic level.⁷ Panel (b) shows that the hiring count through Hello Work remains low, with a small oscillating pattern throughout the period. This stable but limited hiring activity suggests that the platform may face constraints or inefficiencies in increasing the job match rate for part-time workers. In panel (c), the job and worker finding rates (H/U and H/V) exhibit a gradual decline over time, suggesting reduced effectiveness in matching job seekers with available vacancies via Hello Work. This decline might reflect a growing preference for alternative job search methods such as the spot-work platform.

⁷ The decline in vacancies and labor market tightness in the first half of 2020 is consistent with the substantial disruption to labor demand caused by the COVID-19 pandemic in Japan (Fukui and Kikuchi, 2020; Kawaguchi and Motegi, 2021; Kikuchi et al., 2021; Fukai et al., 2021; Ando et al., 2023; Higashi and Sasaki, 2025). This period coincided with a sharp deterioration in economic activity, particularly in face-to-face service sectors where part-time employment is prevalent.

In contrast, the right panels reflect the dynamics of the private platform, where a distinctive pattern emerges. Unlike Hello Work's part-time flow from unemployed to employed, workers on this platform do not automatically leave after securing a match, which keeps a larger pool of active users. The number of registered users increases gradually, while the number of vacancies rises sharply, especially post-2022, leading to a notable rise in labor market tightness (V/U). This reflects an expanding demand for spot employment opportunities. The hiring count in panel (b) shows a rapid upward trajectory starting around 2022, underscoring a marked increase in successful matches.

Panel (c) illustrates the job and worker finding rates (H/U and H/V), highlighting significant differences compared to Hello Work. The job finding rate (H/U) continues to rise, reaching values close to 1.3, indicating that each worker matches with multiple vacancies per month. On the other hand, the worker finding rate (H/V) remains stable around 0.8, suggesting that 80% of the available jobs are successfully filled. This pattern demonstrates the platform's increasing efficiency in matching workers to vacancies. It indicates a maturing spot labor market that still exhibits strong growth.

3. Model

3.1. Nonparametric aggregate matching function

Our primary focus is on analyzing matching efficiency and matching elasticity in relation to the number of registered workers and available vacancies in the labor market, facilitated by an online spot work matching platform operated by Timee in Japan. A matching function derived from search models is central to labor economics.⁸ This function is based on the premise of random search from both sides of the labor market, where job seekers represent labor supply and firms posting vacancies represent labor demand.

To estimate the matching function and recover matching efficiency, we adopt the novel approach proposed by Lange and Papageorgiou (2020).⁹ This approach addresses two critical issues: the endogeneity of matching efficiency (Borowczyk-Martins et al., 2013) and the limitations of the Cobb–Douglas specification, which assumes constant matching elasticity. To overcome these challenges, Lange and Papageorgiou (2020) propose a nonparametric identification and estimation framework for matching efficiency, under specific conditions that will be explored later in this paper.

Let (A, U, V) denote random variables while realizations are subscripted by time t . We define that matching function $m_t(\cdot, \cdot)$ maps period- t unemployed workers U_t , per-capita search efficacy/matching efficiency of the unemployed workers A_t , and vacancies V_t into hires H_t . Let $G(H, U, V)$ and $F(A, U)$ denote the joint distribution of (H, U, V) and (A, U) . Let us assume that V and A are independent conditional on U , that is $V \perp A | U$. Let also assume that the matching function $m(AU, V) : \mathbb{R}_+^2 \rightarrow \mathbb{R}_+$ has constant returns to scale (CRS). Then, by applying nonparametric identification results of Matzkin (2003), Proposition 1 of Lange and Papageorgiou (2020) proves that $G(H, U, V)$ identifies $F(A, U)$ and $m(AU, V) : \mathbb{R}_+^2 \rightarrow \mathbb{R}_+$ up to a normalization of A at one point denoted as A_0 of the support of (A, U, V) .

3.2. Potential limitation

The first limitation of our approach is that the matching efficiency term A may reflect endogenous variation in search effort or hiring standards. As noted in previous work, firms may adjust recruiting intensity and acceptance thresholds in response to labor market conditions, while compositional mismatch can also affect aggregate matching rates.¹⁰ These mechanisms could induce a correlation between A and V , violating the first identification assumption. Although the spot-work setting may attenuate such effects—since employers generally do not engage in selection or screening, and job tasks are relatively homogeneous compared to standard full-time jobs, thereby limiting occupational composition variation—we acknowledge this as a potential source of bias.

Another limitation concerns the interpretation of matching efficiency and elasticity in a gig work environment that differs from traditional unemployment-to-employment models. Platform users are not necessarily unemployed and can take multiple jobs within a period without exiting the market. While this does not pose a problem for estimating matching functions per se—as in the Cobb–Douglas matching function on the TaskRabbit gig work platform in Cullen

and Farronato (2021)—it complicates macroeconomic interpretation of finding rates and elasticities (Wolthoff, 2014). Additionally, although we impose constant returns to scale (CRS) as is standard, online platforms may exhibit increasing returns due to network externalities. In such cases, the estimated efficiency term may also capture scale effects, not purely frictions, altering its structural interpretation (Petrongo and Pissarides, 2001).

Overall, while the estimation remains valid, care must be taken when using the estimated elasticities and efficiency measures as inputs in macroeconomic models or when interpreting them in the context of standard labor market dynamics.

4. Estimation

Following (Lange and Papageorgiou, 2020), we begin by estimating $F(A_0|U)$ across the support of U . To achieve this, we use the distribution of hires conditional on users, U , and observed vacancies, V . Specifically, we have:

$$F(A_0|\psi U_0) = G_{H|U,V}(\psi H_0|\psi U_0, \psi V_0) \quad \text{for any arbitrary scalar } \psi,$$

$F(\psi A_0|\lambda U_0) = G_{H|U,V}(\psi H_0|\lambda U_0, \psi V_0)$ where $\lambda > 0$ is a scaling factor, where $F(A_0|\psi U_0)$ and $G_{H|U,V}$ represent the respective conditional distributions. By varying the parameters (ψ, λ) , we can trace out $F(A|U)$ across the entire support of (A, U) .

Given that our data is finite, we rely on an estimate of $G_{H|U,V}$ for the constructive estimator. Consider an arbitrary point (H_τ, U_τ, V_τ) . To obtain $G(H_\tau|U_\tau, V_\tau)$, we calculate the proportion of observations with fewer hires than H_τ , taken from observations proximate to (U_τ, V_τ) in the (U, V) -space. In practice, this is done by averaging across all observations, assigning smaller weights to those with values (U_t, V_t) distant from (U_τ, V_τ) via a kernel that discounts distant observations. The resulting estimate is expressed as:

$$F(\psi A_0|\lambda U_0) = G_{H|U,V}(\psi H_0|\lambda U_0, \psi V_0),$$

$$\hat{F}(\psi A_0|\lambda U_0) = \sum 1(H_t < \psi H_0) \kappa(U_t, V_t, \lambda U_0, \psi V_0),$$

where $\kappa(\cdot)$ denotes a bivariate normal kernel with bandwidth 0.01.

Once the distribution function $F(A|U)$ is recovered, we invert $F(A_t|U_t)$ to derive A_t for all observations in the dataset, using:

$$A_t = F^{-1}(G(H_t|U_t, V_t)|U_t),$$

Finally, we recover the matching function as:

$$m(A_t U_t, V_t) = G^{-1}(F(A_t|U_t)|U_t).$$

To compute matching elasticities, we employ a linear regression, projecting hires onto the values of vacancies and users, interacted with implied matching efficiency. The resulting estimates approximate the derivatives of the matching function with respect to vacancies and users, interacted with implied matching efficiency. This provides an estimate of the elasticity of the matching function.¹¹

Note that the primary objective of this paper is point estimation of matching efficiency and elasticity. Evaluating sampling uncertainty for these estimates is technically challenging in our setting, because the procedure combines nonparametric conditional distribution estimation for dependent time-series data with subsequent inversion and plug-in steps. For this reason, we focus on point estimates and leave a rigorous assessment of sampling error, such as standard errors or error bands, for future research.

⁸ See Pissarides (2000), Petrongolo and Pissarides (2001), and Rogerson et al. (2005) for further reference.

⁹ Lange and Papageorgiou (2020) also integrate search effort (Mukoyama et al., 2018) and a recruitment intensity index (Davis et al., 2013).

¹⁰ For example, Sedláček (2014) shows that firms' hiring standards can be countercyclical. Similarly, worker search effort (Gomme and Lkhagvasuren, 2015) and firm recruiting intensity (Gavazza et al., 2018) vary over the business cycle. Changes in the composition of workers and jobs can affect the aggregate matching rate, appearing as a change in efficiency (Şahin et al., 2014).

¹¹ The matching elasticity with respect to users $\frac{d \log m(AU, V)}{d \log U} = \frac{dm(AU, V)}{dU} \frac{U}{m} = \frac{dm(AU, V)}{dU} \frac{dAU}{dU} \frac{U}{H} = \frac{dm(AU, V)}{dAU} \frac{AU}{H} = \frac{d \log m(AU, V)}{d \log AU}$ is obtained from the regression coefficient of H on AU and multiplying it by $\frac{AU}{H}$.

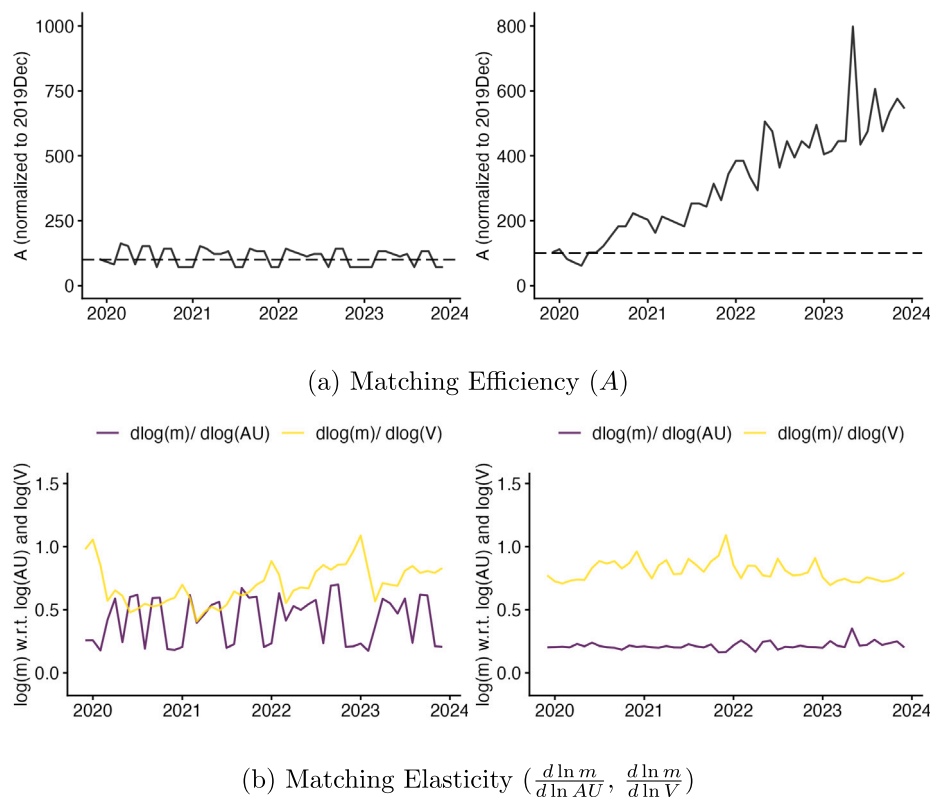


Fig. 4. Hello Work part-time vs platform 2019–2023. Note: For confidentiality reasons, we normalize the efficiency to 2019 December for each platform.

5. Results

5.1. Matching efficiency and elasticity in the platform

Fig. 4 illustrates the trends in matching efficiency, normalized so that December 2019 equals 100, and matching elasticity for both Hello Work (left panels) and the private platform (right panels) from December 2019 to December 2023. In panel (a), the dashed horizontal line indicates this baseline level of 100, corresponding to the normalized matching efficiency in December 2019. The matching efficiency for Hello Work part-time remains relatively stable around the baseline, fluctuating within a narrow range without any significant upward or downward trends. In contrast, the private platform shows a markedly different pattern, with matching efficiency experiencing sharp increases, peaking around May 2023 at around 800, which implies that the efficiency is about eight times larger than in the initial period in December 2019.

Panel (b) presents the matching elasticities with respect to users and vacancies for both platforms. For Hello Work (left panel), the elasticity with respect to users fluctuates between approximately 0.3 and 0.5, while the elasticity with respect to vacancies ranges roughly from 0.4 to slightly above 1.0, with somewhat higher volatility, particularly in the earlier years. This suggests that the matching process is only moderately responsive to changes in users and vacancies and exhibits limited variation over time.¹²

In contrast, the private platform (right panel) exhibits consistently stable and relatively lower elasticities. The elasticity with respect to

users remains relatively stable, ranging between 0.2 and 0.3. The vacancy elasticity is particularly elevated and stable, mostly ranging between 0.7 and 1.1. One caveat in interpreting this difference is that, in the platform data, an active user is counted only once within a month even if the same worker matches with multiple jobs, whereas vacancies are recorded at the posting level. This data structure may mechanically attenuate the estimated user elasticity relative to the vacancy elasticity. Nevertheless, even taking this measurement issue into account, the vacancy-side elasticity is substantially larger than the user-side elasticity throughout the sample period. Therefore, the estimates potentially demonstrate that the platform's matching process is more strongly driven by vacancy-side conditions than by active-user inflows.

5.2. Prefecture-level matching efficiency and elasticity in the platform

Fig. 5 illustrates prefecture-level labor market trends across the three representative prefectures – Aichi, Osaka, and Tokyo – which are the centers of the three metropolitan areas in Japan. This exercise is intended to assess whether the platform exhibits similar matching performance across major regions. We focus on these three prefectures because many smaller prefectures are too thin, especially early in the sample, for spot-work matching to form a meaningful market-level series, and because (Kanayama et al., 2026) shows that Timee first launched in major metropolitan prefectures such as Tokyo, Osaka, and Aichi. For confidentiality reasons, the y-axis in the following figures reflects values relative to the baseline set by Tokyo in December 2019. Panels (a) and (b) display user and vacancy levels. Tokyo consistently leads in both categories, with particularly steep growth in vacancy postings after 2022, reflecting rapid labor demand expansion. Panel (c) shows that hires increase in tandem, especially in Tokyo, but Aichi and Osaka also experience steady growth.

¹² The estimated elasticities differ from those reported in Otani (2024), likely due to the differing lengths of time horizons considered. Specifically, Otani (2024) includes data from 1972 to 2013, a period covering economic booms and busts, which captures a broader range of labor market dynamics.

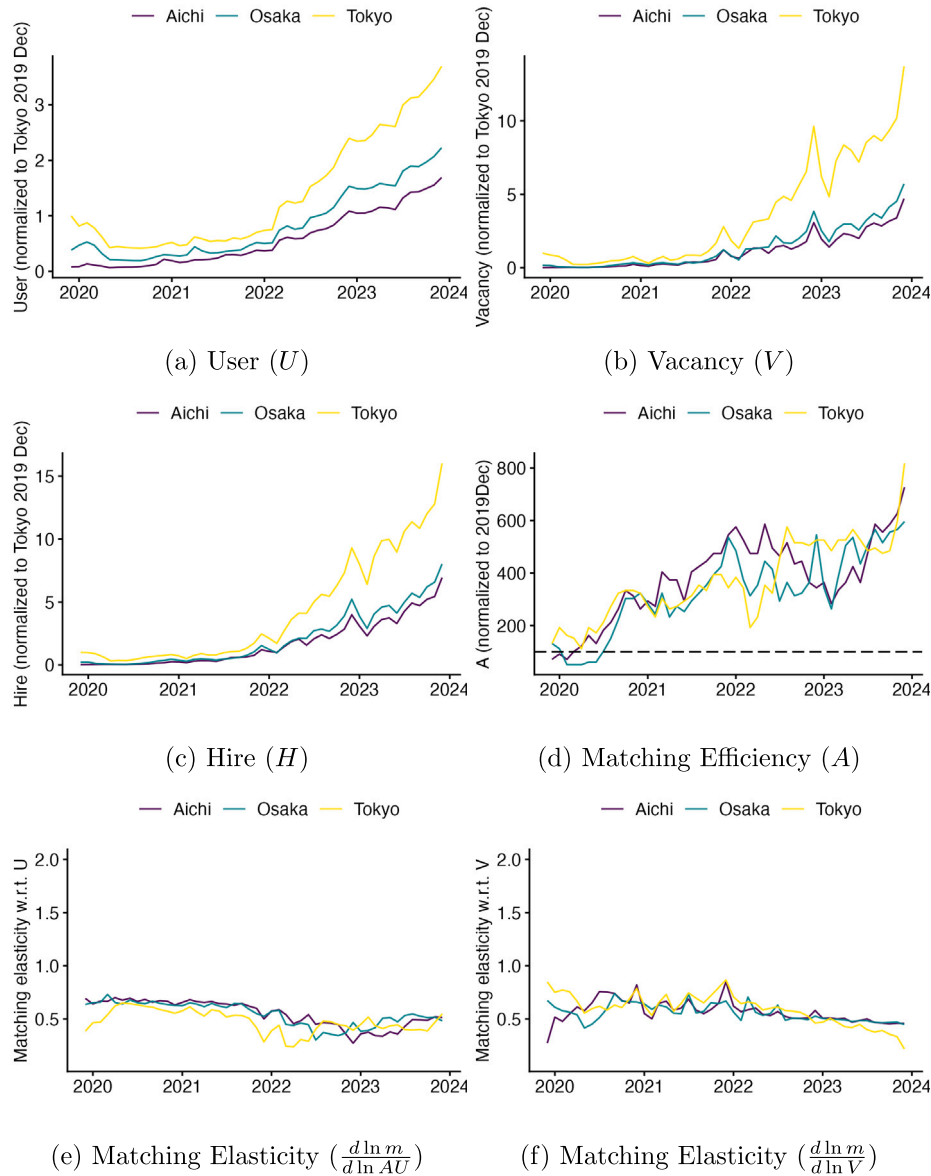


Fig. 5. The heterogeneity across prefectures 2019–2023.

Note: For confidentiality reasons, we normalize the level of the y-axis in Panels (a)–(c) to the level of Tokyo in December 2019. In Panel (d), matching efficiency for all three prefectures is normalized by the Tokyo value in December 2019, which is set equal to 100. The dashed horizontal line denotes this common Tokyo-based benchmark.

Panel (d) presents trends in matching efficiency across the three prefectures. Matching efficiency for all three prefectures is normalized by the Tokyo value in December 2019, which is set equal to 100. The dashed horizontal line denotes this common Tokyo-based benchmark. We use this common baseline to facilitate comparisons across prefectures. The three prefectures show broadly similar movements from 2020 to 2021, followed by a period of divergence from 2021 to 2023. After 2023, the gap in matching efficiency narrows somewhat, indicating partial re-convergence, although noticeable differences remain near the end of the sample. This suggests that regional heterogeneity in platform matching efficiency became less pronounced later in the sample, but was not fully eliminated. Panels (e) and (f) report elasticities with respect to users and vacancies, respectively. Both elasticities remain stable across time and across the three prefectures, generally ranging from 0.3 to 0.8 for user elasticity and from 0.4 to 0.6 for vacancy elasticity. The absence of persistent cross-prefecture gaps indicates limited geographic heterogeneity in matching responsiveness on the platform.

6. Conclusion

This study reveals significant differences in labor market dynamics between the private online spot work platform and Hello Work, Japan's public employment service. The private platform demonstrates a rapid expansion in part-time spot employment from December 2019 to December 2023, with a notable increase in registered users and a sharp rise in vacancies, particularly after 2022. We find that the private platform has become increasingly effective in facilitating job matches, achieving a worker finding rate where 80% of spot jobs are filled, reflecting its growing influence in the part-time labor market.

By employing a novel nonparametric approach, the study estimates the matching function, highlighting clear distinctions in matching efficiency and elasticity between the two platforms. For Hello Work, matching efficiency remains stable around the baseline, while the elasticity with respect to users fluctuates around 0.3–0.5 and the elasticity with respect to vacancies ranges roughly from 0.4 to slightly above 1.0. In contrast, the private platform exhibits higher volatility, with

matching efficiency peaking in 2023. The elasticity with respect to the number of users fluctuates between 0.2 and 0.3, while the elasticity concerning vacancies fluctuates between 0.7 and 1.1. These patterns suggest that the private platform's matching process is more vacancy-driven and systematically more responsive to job demand fluctuations than the public counterpart.

At the prefecture level, the three prefectures exhibit broadly similar movements early in the sample, followed by divergence and partial re-convergence later on. Both user- and vacancy-side elasticities remain stable and close in magnitude across regions, pointing to limited geographic heterogeneity in matching responsiveness on the platform.

This study offers macroeconomic implications and serves as a foundational step toward understanding labor market dynamics at the micro-level. More broadly, the comparison between Hello Work and the platform provides useful insight for modeling gig spot work within search-and-matching frameworks, particularly by highlighting how repeated matching, app-based search, and differences in measurement shape observed matching outcomes. Future research will explore the behaviors of individuals on both the labor demand and supply sides and their interactions that drive aggregate patterns. This approach will enable a more detailed analysis of the mechanisms underpinning labor market functioning.

CRedit authorship contribution statement

Hayato Kanayama: Writing – original draft, Validation, Project administration, Investigation, Data curation. **Suguru Otani:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Project administration, Methodology, Data curation, Conceptualization.

Data availability

The data that has been used is confidential.

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